



NORTHERN UNIVERSITY

OF BUSINESS & TECHNOLOGY KHULNA

Department of Computer Science and Engineering

Project documentation

Course Title: Mobile Computing LAB

Course Code: CSE 4204

Project name: Medical Report Analyzer

Submitted To:

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Submitted By:

Team : 02

Section: 8D

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Team Information

Team Name	CSE4204-8D-T02
Section	8D
Project Title	Medical Report Analyzer

Team Members & Task Distribution

Student ID	Name	Role	Task Distribution
11220320899	Farjana Easmin Minka	AI Integration Lead & Database Manager	AI Integration
11220320959	Hassan A.K Azad Tushin	Team Leader	Documentation, Database Management, Testing
11220320982	Juena Tabassum	Frontend Developer	Frontend Development: React.js
11220321050	Faika Newaz Silmi	Backend Developer	Backend Development: Node.js, Express.js

GitHub Repository Link <https://github.com/Tushin-vai/CSE-4204-8D-T02>

Medical Report Analyzer AI Assistant

1.1 Project Overview

The Medical Report Analyzer is an AI-powered web application designed to bridge the gap between complex medical documentation and patient comprehension. By leveraging large language models, the system translates clinical terminology, lab values, diagnoses, and prescriptions into plain, readable language that any patient can understand without a medical background.

1.2. Problem Statement

Medical reports are filled with clinical terminology, abbreviations, and technical jargon that is difficult for the average person to understand. Patients receiving blood test results, discharge summaries, lab reports, or prescriptions are often left confused about what the findings mean and what actions they should take.

Why the problem matters:

- Millions of patients receive medical reports each year without the ability to understand them fully.
- Misunderstanding a report can lead to delayed treatment, unnecessary anxiety, or poor health decisions.
- Patients often rely on costly follow-up appointments just to have a doctor explain what the report says.

Why existing solutions are insufficient:

- Generic search engines return broad medical information, not personalized analysis of a specific report.
- Rule-based tools cannot handle the variability in report formats, abbreviations, and layouts across hospitals and labs.
- No current mainstream tool offers a conversational, interactive way to ask questions about your own report.

Why AI is relevant:

- Large language models can read unstructured medical text, understand clinical context, and rephrase findings in plain language.
- Conversational AI enables a natural Q&A experience that static tools simply cannot replicate.

1.3 Objectives

- Provide patients with instant, plain-language summaries of their medical reports.
- Extract and highlight key findings such as diagnoses, lab values, and abnormal results.
- Offer an interactive chatbot for follow-up questions grounded in the uploaded report.
- Securely store reports, summaries, and chat histories per authenticated user account.
- Support both text-pasted and image/scanned report uploads via OpenAI Vision.

1.4 Scope

The system is scoped to serve individual patients who receive medical documents from hospitals, clinics, or diagnostic laboratories. The application will:

- Accept medical reports as typed text, PDF files, or scanned images.
- Generate AI-powered summaries and structured key-finding extractions.
- Provide a conversational chatbot interface to answer report-specific questions.
- Maintain a per-user report history accessible at any time.

Out of scope: direct integration with hospital EHR systems, real-time diagnostic advice, and multi-language support in version 1.0.

Core Features

- **Instant Summaries:** Converts complex health documents into simple, easy-to-read overviews targeted at an 8th-grade reading level or below.
- **Smart Extractions:** Automatically pulls out key findings like medications and lab values, visually highlighting abnormal results in red so they instantly stand out.

- **Interactive Chatbot:** Features a built-in AI assistant that lets patients ask natural follow-up questions directly grounded in their specific report.
- **Flexible Uploads:** Accepts reports via typed text, PDF files, or scanned photos/images.
- **Secure History:** Securely saves past reports and chat histories under a private user account so patients can review them anytime.

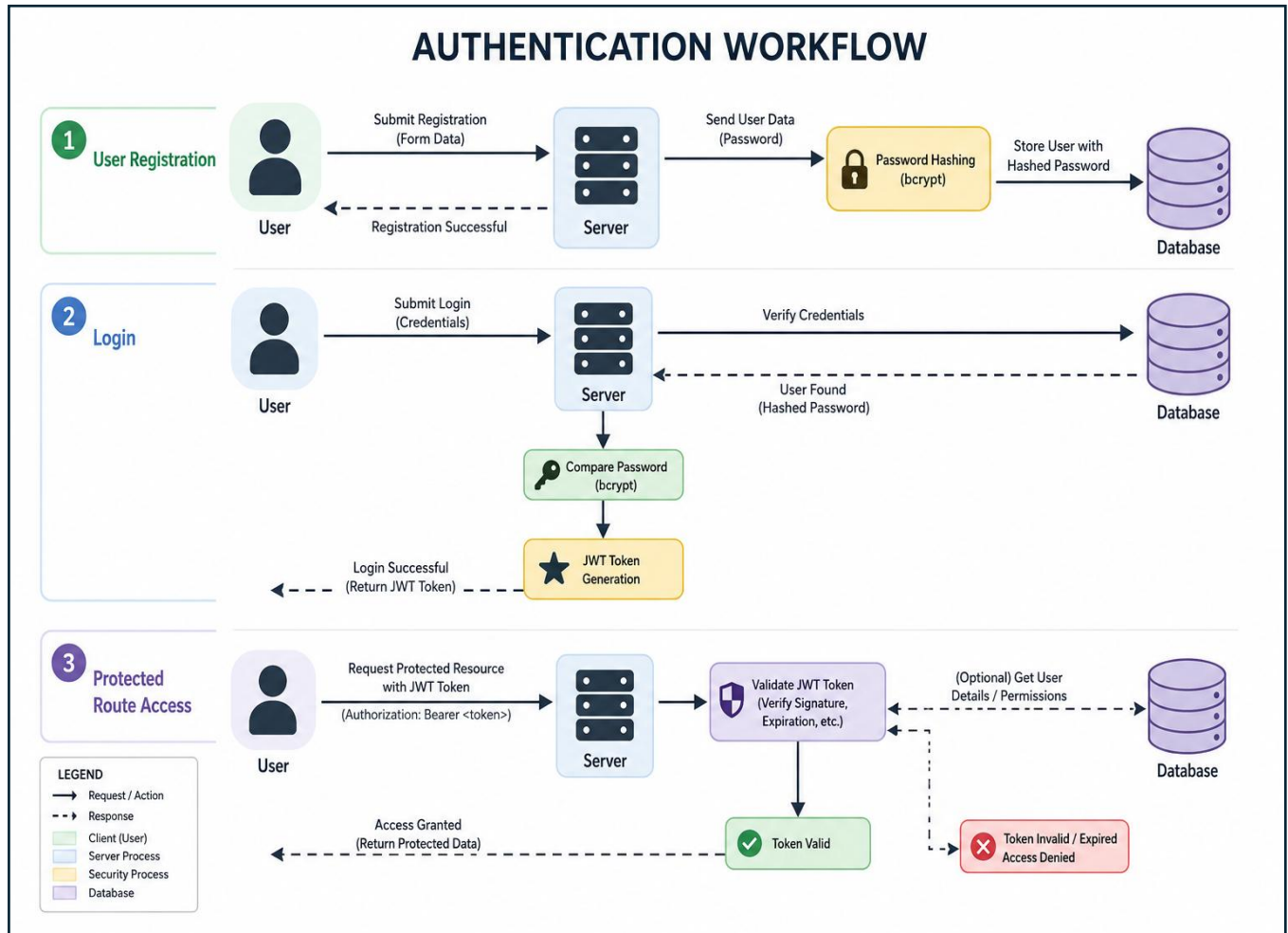
Constraints & Scope

- **Informational Only:** The app is strictly an educational tool to help patients understand their data; it does not provide real-time clinical diagnoses or treatment advice.
- **Privacy First:** The platform is designed with data protection (HIPAA compliance) in mind. Administrators can monitor system health, but they are strictly locked out of viewing any patient's actual medical content or chat history.
- **Initial Release:** Version 1.0 is a web-based platform optimized for English-language reports.

3. Backend Technology Stack

Category	Technology	Purpose
Runtime	Node.js v24	JavaScript server-side runtime
Framework	Express.js v4.19	REST API framework
Database	PostgreSQL (via Supabase)	Relational database, cloud-hosted
Authentication	JWT (jsonwebtoken v9)	Access + Refresh token auth
Password Security	bcryptjs v2.4 (12 rounds)	Secure password hashing
AI Integration	Mocked (OpenAI GPT-4 ready)	Medical report analysis & chat

Authentication Workflow :



7. Current Development Progress

#	Task	Status	Details
1	Backend Project Setup	☑ Completed	Node.js + Express, folder structure, all dependencies
2	Database Implementation	☑ Completed	5 tables, PKs, FKs, constraints, indexes on Supabase
3	Authentication System	☑ Completed	JWT access+refresh tokens, bcrypt, role management
4	Core API Development	☑ Completed	13 endpoints across Auth, Profile, Reports, Chat
5	Database Connectivity	☑ Completed	Supabase PostgreSQL connected, all CRUD operations working
6	Error Handling	☑ Completed	Global handler, PostgreSQL error codes, JWT errors, 404
7	API Testing	☑ Completed	Postman collection with 21 test cases, auto token chaining

Screensorts of API Testing:

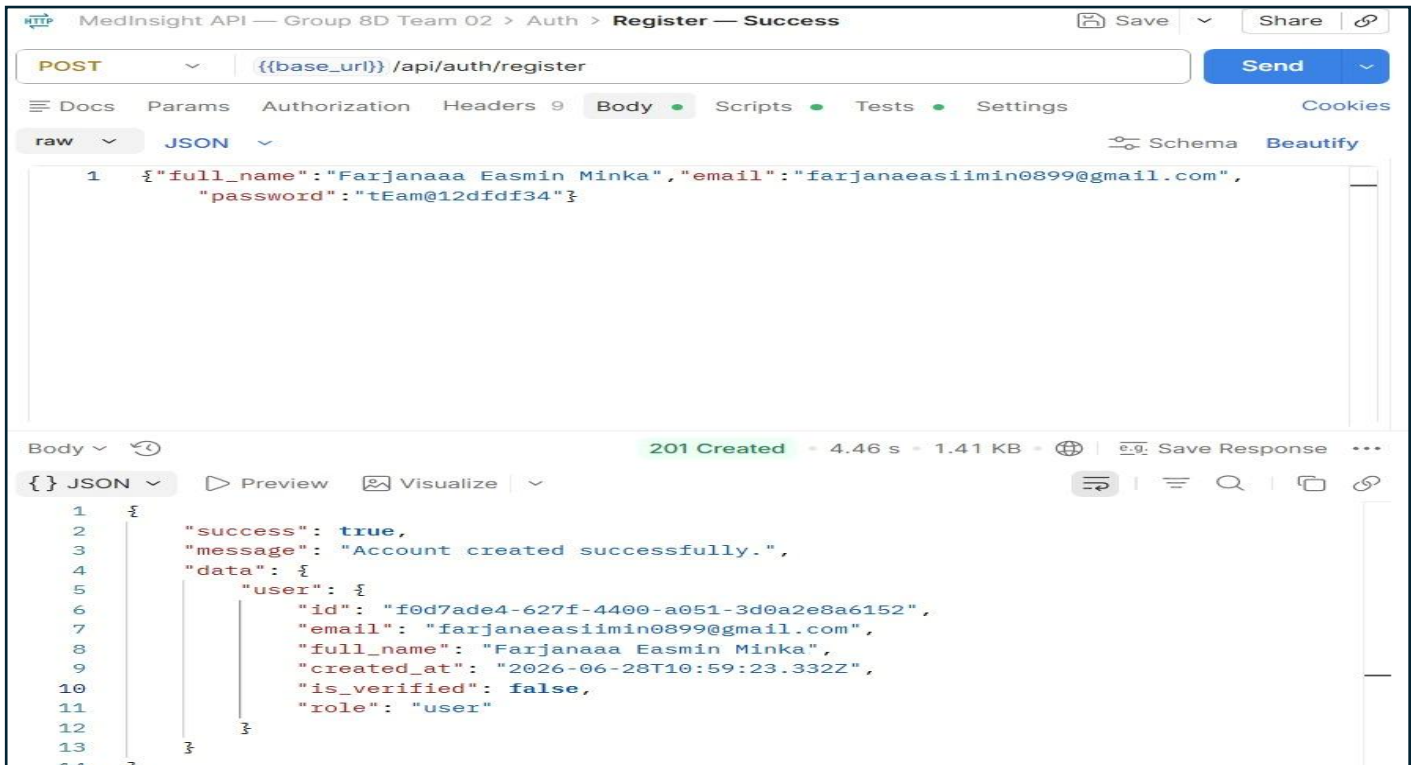


Fig : Resgistretion Demo

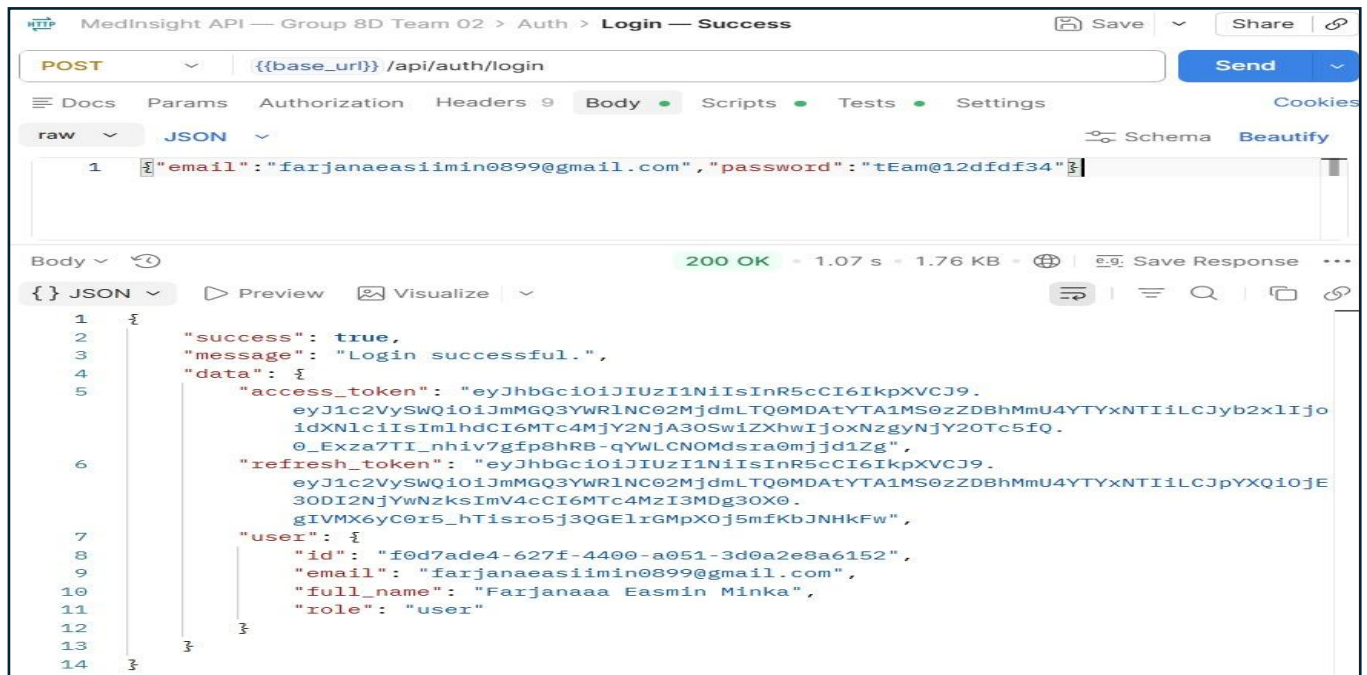


Fig :Log in Demo

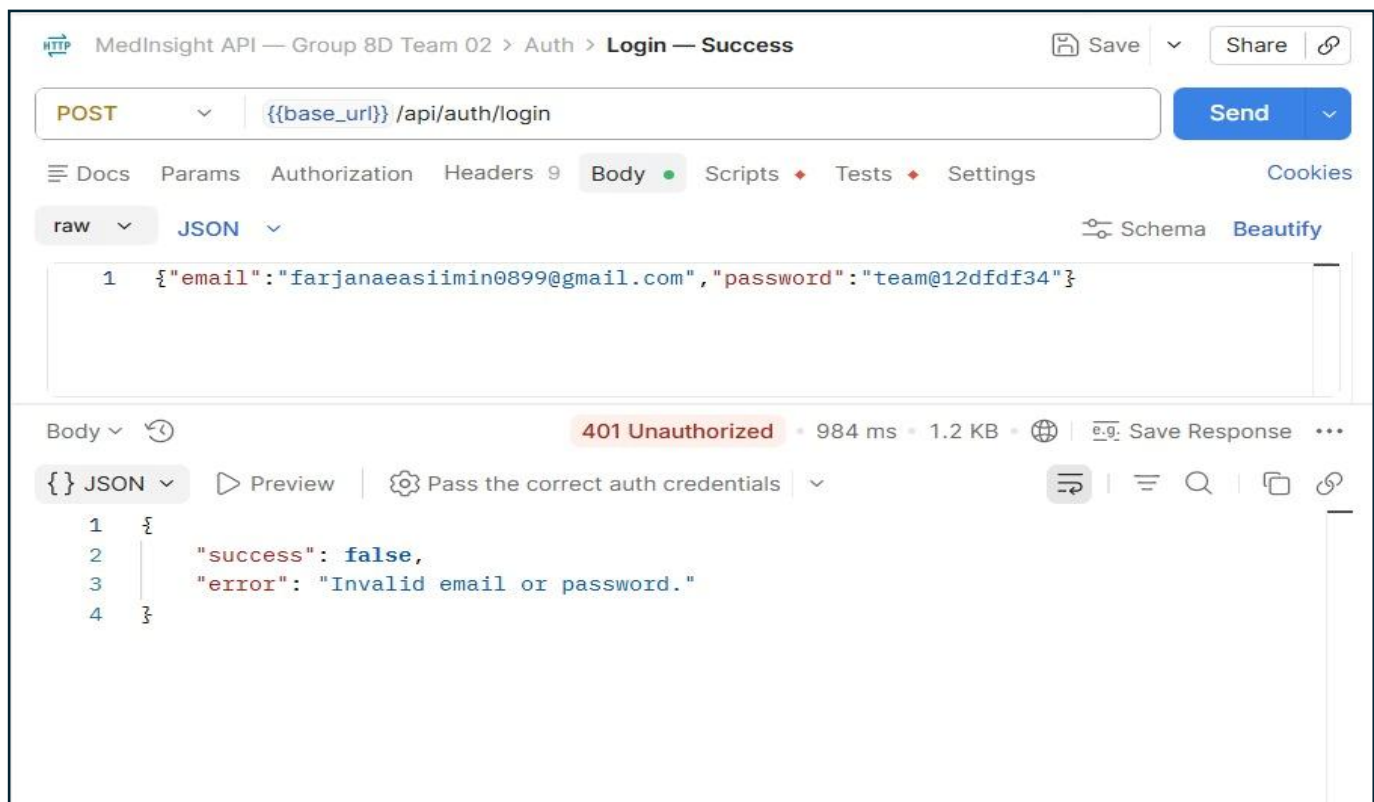


Fig: Authentication Error Identification

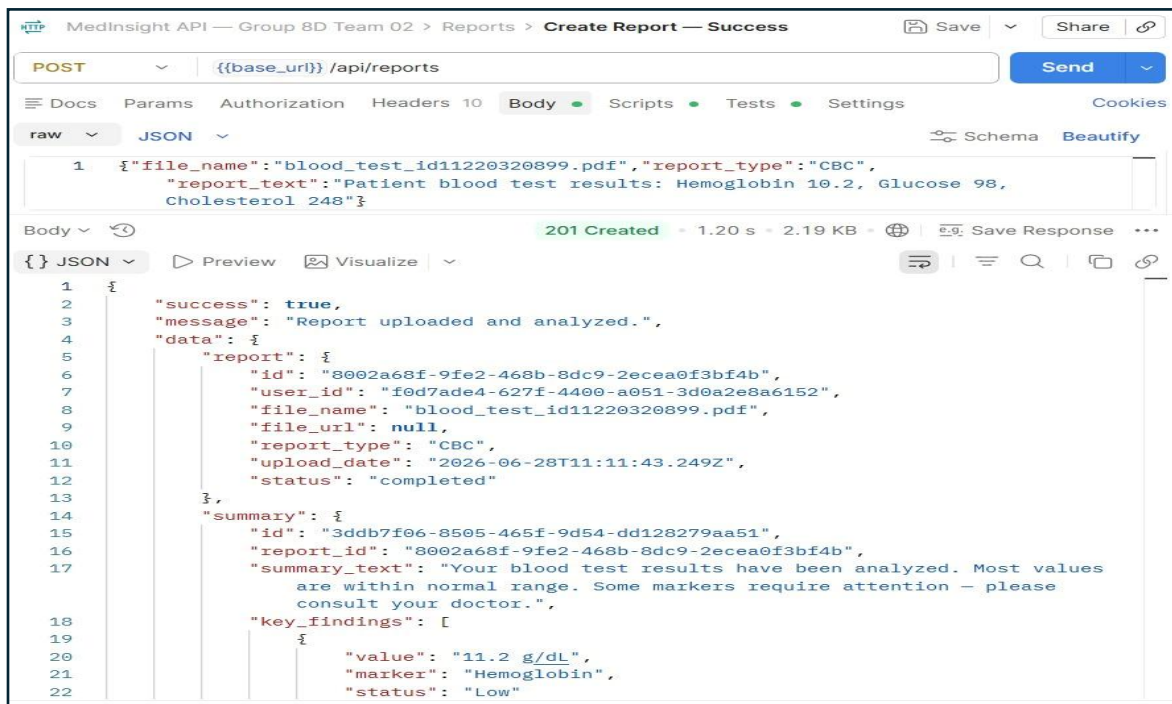


Fig: Text report analyzer

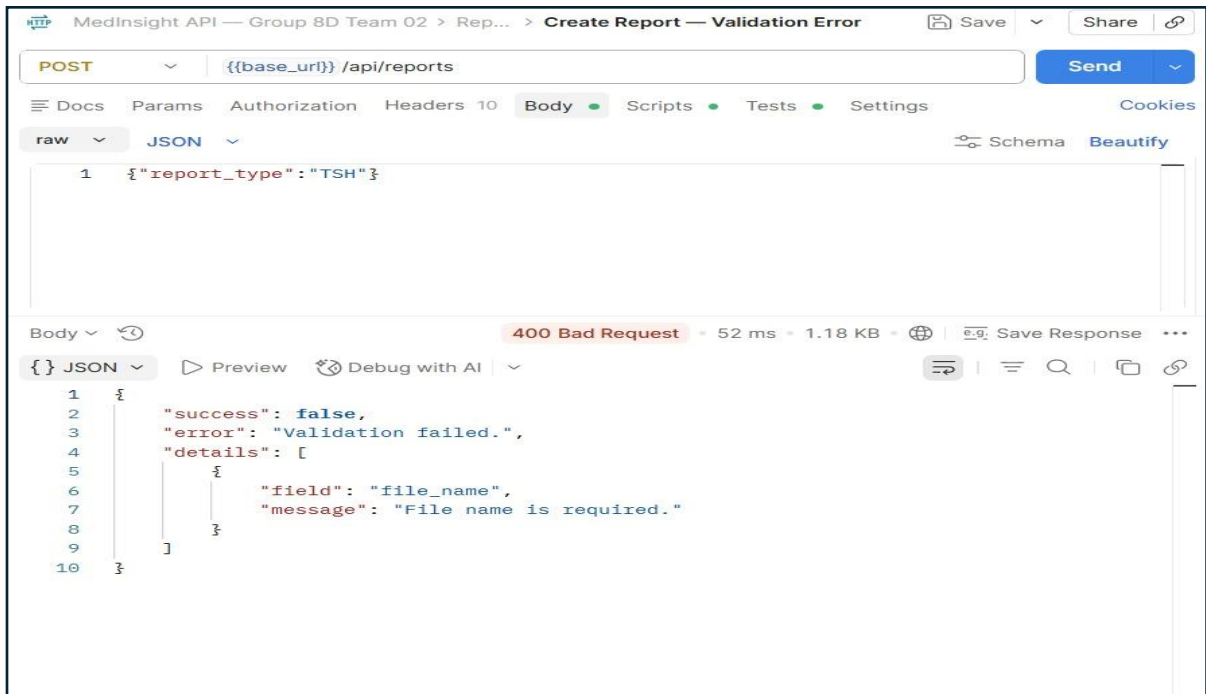


Fig: Error Handling

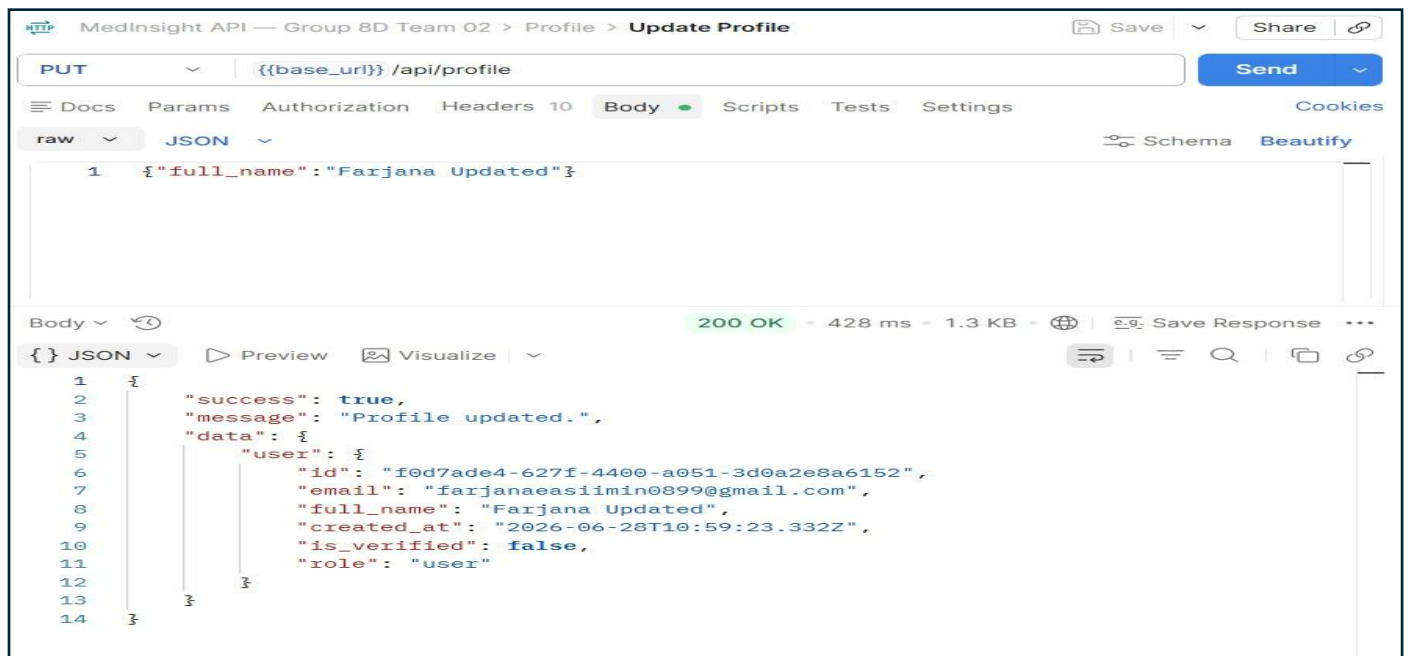


Fig : Updated Profile

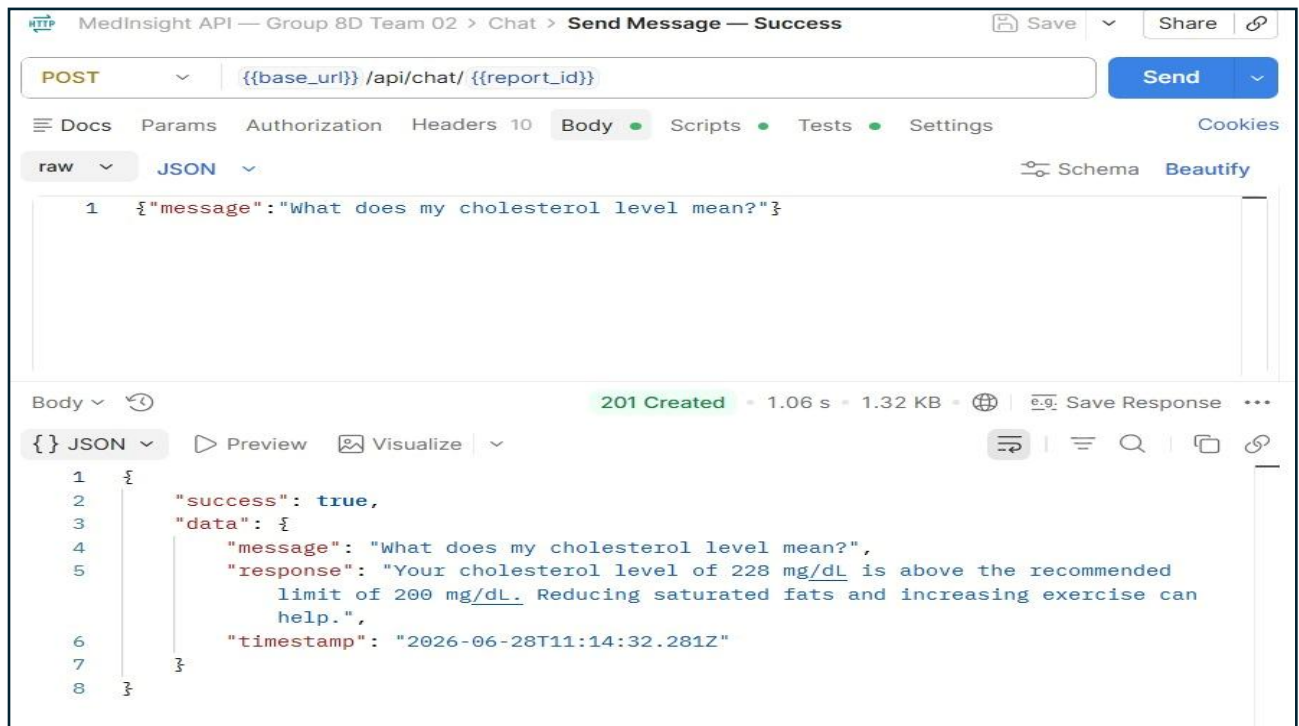


Fig : Response user Query

Table Editor	users	sessions	reports	chat_history	+
schema public + New table Search tables... chat_history reports sessions summaries users	Filter by id, email, password_hash... or ask AI Sort Add RLS policy Role postgres Insert				
	id	email	password_hash	full_name	created_at
	4fb30897-1557-4bd3-86ed-6edf44e508de	farjanaeasmin@gmail.com	\$2a\$12\$ETWaP1jxaTBcIW7xl8ORGU5b	Farjana Test	2026-06-27 16:06:41.91E
	89157ea5-8368-4b99-a7d9-2f038d6d0e5f	farjana@test.com	\$2a\$12\$TGTOFp1Cu/3g8wcvjFqB5.VUL2n	Farjana Test	2026-06-26 21:17:52.632E
	99eed98c-95f9-4f6b-8d32-c5b1bcbe0d6f	farjanaeasiimin899@gmail.com	\$2a\$12\$eeuMTwcA97mg/2gp5u7qW.2zP	Farjanaaa Test	2026-06-28 16:11:54.495E
	d6eateaa-d18d-4cb6-9f51-e566d5dc9fe3	farjanaeasmin89@gmail.com	\$2a\$12\$//WLIE3U4d8HTA8tIUkOUO8TrRI	Farjana Test	2026-06-27 17:49:45.195E
	f0d7ade4-627f-4400-a051-3d0a2e8a6152	farjanaeasiimin0899@gmail.com	\$2a\$12\$QcZocTcm12LhQF.p672bX.HU17g	Farjana Updated	2026-06-28 16:59:23.332E

Fig: Auth Database storage

Table Editor

users

sessions

reports

chat_history

summaries

+

schema public

+ New table

Search tables...

chat_history

reports

sessions

summaries

users

Filter by id, report_id, summary_text... or ask AI

Sort

Add RLS policy

Role postgres

Insert

	id uuid	report_id uuid	summary_text text	key_findings jsonb	abnormal_flags jsonb
	3ddb7f06-8505-465f-9d54-dd128279aa51	8002a68f-9fe2-468b-8dc9-2ecea...	→ Your blood test results have been analyze	{("value":"11.2 g/dL","marker":"Hemoglobi	{("marker":"Hemoglobin'
	8c1bc88d-227f-441a-9fe0-100f94cb2a55	db1c3be4-1792-4cf2-a3ce-8bf04d...	→ Your blood test results have been analyze	{("value":"11.2 g/dL","marker":"Hemoglobi	{("marker":"Hemoglobin'
	8daf1f8d-8738-4f52-bcb2-59e5a44cedea	90e76e1a-c1e7-461b-bb05-190c3b...	→ Your blood test results have been analyze	{("value":"11.2 g/dL","marker":"Hemoglobi	{("marker":"Hemoglobin'

Fig : Summarize report store in database

Table Editor

users

sessions

reports

chat_history

summaries

+

schema public

Switch schema S then S

Search tables...

chat_history

reports

sessions

summaries

users

Filter by id, user_id, file_name... or ask AI

Sort

Add RLS policy

Role postgres

:

↺

Insert

	id uuid	user_id uuid	file_name varchar	file_url text	report_type varchar
	8002a68f-9fe2-468b-8dc9-2ecea0f3bf4b	f0d7ade4-627f-4400-a051-3d0a2...	blood_test_id11220320899.pdf	NULL	CBC
	90e76e1a-c1e7-461b-bb05-190c3b0db841	f0d7ade4-627f-4400-a051-3d0a2...	blood_test.pdf	NULL	CBC
	db1c3be4-1792-4cf2-a3ce-8bf04db76757	89157ea5-8368-4b99-a7d9-2f038...	blood_test.pdf	NULL	CBC

Fig : Report Store in database

Table Editor		users	sessions	reports	chat_history	summaries		
schema public		Filter by id, report_id, user_id... or ask AI					Sort	Add RLS policy
+ New table							Role postgres	Insert
Search tables...								
chat_history								
reports								
sessions								
summaries								
users								

id	report_id	user_id	message	response
f72960-36da-4de8-a1ba-1cdda4a1a824	db1c3be4-1792-4cf2-a3ce-8bf04d...	89157ea5-8368-4b99-a7d9-2f038...	What does my cholesterol level mean?	Your cholesterol level of 228
c73d2a-d16b-448c-a2d2-6c99cbc26fd1	8002a68f-9fe2-468b-8dc9-2ecea...	f0d7ade4-627f-4400-a051-3d0a2...	do i have thyroid?	Based on your report, your r
784e8c-7d44-4e61-9cf3-829a9f9b8a54	8002a68f-9fe2-468b-8dc9-2ecea...	f0d7ade4-627f-4400-a051-3d0a2...	What does my cholesterol level mean?	Your cholesterol level of 228

Fig: Chat history store in database

Backend Folder Structure

